

LOGY TECH COMPANY

NETWORK IMPLEMENTATION PROJECT REPORT

1. Ezz Aldeen Abdulfattah Abdoh Abdulsalam Alshalafi, Cyber Security Student, University of Science and Technology, Sana'a, G1.
Corresponding ID: 202310101160

2. Professor, University of Science and Technology, Sana'a — Dr. Ameen.

- **Project Mandate**

This document details a comprehensive plan for the design, configuration, and implementation of a secure and scalable network infrastructure. The project's core objective is to apply a robust theoretical understanding of Cisco Certified Network Associate (CCNA) principles to solve the real-world networking challenges of a multi-branch corporate entity.

- **Project And Corporate Synopsis**

- **Project Sponsor:** Logy Tech
- **Locations:** in Dubai City in two areas; Downtown and Business Bay.

- **Date of Submission:** Aug 18th, 2025

Lab Title: Network Implementation Project

First: Description

This lab demonstrates the design and configuration of a secure, multi-department enterprise network for Logy Tech Company across its two locations Area in Dubai City. The implementation focuses on VLAN segmentation for departmental isolation, DHCP, inter-VLAN routing, OSPF dynamic routing between sites, and static routing where necessary. Security is enforced through comprehensive ACL policies and enhanced with port security, and internet access is provided via PAT.

Second: Design

1. Topology

The new network adopts the same **hierarchical three-tier model** (core–distribution–access) that was proven in the template, but is now sized and cabled for the **Downtown** facility.

- **Core Layer:** Two redundant routers (Core-R1, Core-R2) and two multilayer switches (MLSW-1, MLSW-2) form the resilient backbone.
- **Distribution Layer:** Each floor in Each Area is served by a dedicated distribution switch (Multilayer Switch -1, Multilayer Witch -2) aggregates departmental traffic.
- **Access Layer:** In Each Department in floors consists of End-user PCs, printers<except the HR Department>, and Cisco Aironet APs connect to the access ports of the distribution switches.

2. Components;

The network design for the project incorporates the following devices:

- Routers (4)
 - 2 ISP router for upstream connectivity.
 - Positioned at the core layer for redundancy.
 - Connect to both ISPs for internet connectivity.
 - Configured with static, public IP addresses from ISPs.
- Multilayer Switches (2)
 - Deployed at the core layer to provide redundancy and efficient routing.
 - Configured for both switching and routing functionalities.
 - Assigned IP addresses to enable inter-VLAN routing.
- Distribution Layer Switches (Multiple)
 - Connect individual departments to the core layer.
 - Facilitate communication within respective VLANs.

- End-User Devices (PCs)
 - Deployed at the access layer.
 - Connected to distribution layer switches for departmental access.
- Cisco Access Points (APs)
 - Positioned at the access layer to provide wireless connectivity.
 - Ensure wireless network availability in each department.
- DHCP Servers (1)
 - Located in the server room.
 - Dynamically allocate IP addresses to end-user devices.
- Server Room Devices (Servers, etc.)
 - DNS server, HTTP server etc.
 - Devices in the server room are allocated static IP addresses.
 - These devices may include servers, storage units, and networking equipment.

3. IP Addressing Scheme

Provide details about the IP addressing scheme applied to the network in each Area;

Base Network: 192.168.0.0/22

Downtown;

- First floor

Devices	Network Address	Subnet Mask	Usable Host Range	Gateway
Sales & Marketing (VLAN 10)	192.168.10.0/24	255.255.255.0	192.168.10.1-192.168.10.254	192.168.10.1
Sales_PC, Sales-Printer, Sales-Laptop, Sales-Tablet	DHCP	----	----	----
HR (VLAN 20)	192.168.20.0/24	255.255.255.0	192.168.20.1 - 192.168.20.254	192.168.20.1
HR-PC, HR-Laptop, HR-Tablet	DHCP	255.255.255.0	--	----

- Second Floor

Devices	Network Address	Subnet Mask	Usable Host Range	Gateway
Finance &Accounts (VLAN 30)	192.168.30.0/24	255.255.255.0	192.168.30.1 -192.168.30.254	192.168.30.1
Finance-PC, Finance-Printer, Finance-Laptop, Finance-Tablet	DHCP	--	--	---
Admin & Public Relations (VLAN 40)	192.168.40.0/24	255.255.255.0	192.168.40.1 -192.168.40.254	---
Admin-PC, Admin-Printer, Admin-Laptop, Admin-Tablet	DHCP	----	----	----

Business Bay;

Devices	Network Address	Subnet Mask	Host Address Range<device>	Default Gateway
ICT (VLAN 50)	192.168.50.0/24	255.255.255.0	192.168.50.1-192.168.50.254	192.168.50.1
ICT-PC, ICT-Printer, ICT-Laptop, ICT-Tablet	DHCP	---	----	----
Servers (VLAN 60)	192.168.60.0/24	255.255.255.0	192.168.60.1--192.168.60.254	192.168.60.1
DHCP-Server	192.168.60.2	---	---	192.168.60.1
DNS-Server	192.168.60.3	---	---	----
SysAdmin-PC	192.168.60.4	----	----	----
Email-Server	192.168.60.5	----	---	----

L3 Multilayer Switches Links (Distribution Layer);

Mlt-SW1

Interface	Network Address	Subnet Mask	Host Address Range<H>	Connection
Gig1/0/1	10.10.10.2/30	255.255.255.252	10.10.10.1 - 10.10.10.2	CORE-R1
Gig1/0/2	10.10.10.10/30	255.255.255.252	10.10.10.9 - 10.10.10.10	CORE-R2
VLAN (10)	192.168.10.1/24	255.255.255.0	192.168.10.1 –192.168.10.254	Sales &Market gate
VLAN (20)	192.168.20.1/24	---	192.168.20.1 –192.168.20.254	HR gate
VLAN (30)	192.168.30.1/24	----	192.168.30.1 –192.168.30.254	Finance gate
VLAN (40)	192.168.40.1/24	----	192.168.40.1 –192.168.40.254	Admin & PR gate
VLAN (50)	192.168.50.1/24	----	192.168.50.1 –192.168.50.254	ICT gate
VLAN (60)	192.168.60.1/24	---	192.168.60.1 –192.168.60.254	Server Room
Gig1/0/1	10.10.10.6/30	255.255.255.252	10.10.10.5 - 10.10.10.6	CORE-R1
Gig1/0/2	10.10.10.14/30	---	10.10.10.13 - 10.10.10.14	CORE-R2
VLAN (10)	192.168.10.1/24	255.255.255.0	192.168.10.1 –192.168.10.254	Sales &Market gate
VLAN (20)	192.168.20.1/24	---	192.168.20.1 –192.168.20.254	HR gate
VLAN (30)	192.168.30.1/24	---	192.168.30.1 –192.168.30.254	Finance gate
VLAN (40)	192.168.40.1/24	---	192.168.40.1 –192.168.40.254	Sales &Market gate
VLAN (50)	192.168.50.1/24	---	192.168.50.1 –192.168.50.254	HR gate
VLAN (60)	192.168.60.1/24	---	192.168.60.1 –192.168.60.254	Server Room gate

Mlt-SW2

Public IP between Core and ISP;

- 103.133.254.0/30
- 103.133.254.4/30
- 103.133.254.8/30
- 103.133.254.12/30

ISP Addressing table;

Devices	Interface	IP Address	Subnet
ISP-1	S0/3/0	103.133.254.1/30	255.255.255.252
	S0/3/1	103.133.254.5/30	---
ISP-2	S0/3/0	103.133.254.9/30	---
	S0/3/1	103.133.254.13/30	----

Core Layer Addressing table;

Devices	Interface	IP Address	Subnet
Core-R1	G0/0	10.10.10.1	255.255.255.252
	G0/1	10.10.10.5	----
	S0/2/0	103.133.254.1	----
	S0/2/1	103.133.254.10	----
Core-R2	G0/0	10.10.10.9	----
	G0/1	10.10.10.13	---
	S0/2/0	103.133.254.6	---
	S0/2/1	103.133.254.14	--

Third: Network Implementation Tasks

Requirements;

- Design a hierarchical topology in Cisco Packet Tracer.
- Connect devices with proper cabling and configure basic settings.
- Create VLANs, assign ports, and configure subnetting with IP addressing.
- Enable inter-VLAN routing on multilayer switches.
- Configure a dedicated DHCP server for dynamic IP allocation.
- Set up SSH for secure remote access.
- Implement OSPF routing protocol.
- Configure NAT overload (PAT).
- Apply standard and extended ACLs.
- Enable switchport (port) security.
- Configure WLAN using Cisco Access Points.
- Configure host devices and ISP routers.
- Test and verify end-to-end network communication.

All of the requirements going to be classified according to the course (CCNA) in steps

Objectives

- Step 1: Routing Configuration.
- Step 2: Switching Configuration.
- Step 3: Inter-VLAN Routing.
- Step 4: Security Measures.
- Step 5: Quality of Service (QoS).
- Step 6: Monitoring and Management.
- Step 7: Testing and Validation
- Step 8: Results and Evaluation (Verifications)
- Security Notes:
 - Switchport security just in finance department.
 - Mitigate serverRoom devices with static IP address
 - Inter-VLAN routing on the 13 switches plus ip dhcp helper addresses.
 - Configure DHCP in Server in serverRoom in one place for detection on place with high Security Tech.

Steps

Step 1: Routing Configuration

- **Basic Router Configuration**
 - **Hostname configuration** – Sets the router's name.
 - **Console and enable password setup** – Secures router access.
 - **Disable IP domain lookup** – Avoids unwanted DNS queries.
 - **MOTD banner configuration** – Shows login message.
 - **Password encryption service** – Encrypts stored passwords.
 - **SSH setup** – Enables secure remote access.
 - **User creation** – Adds local user

===== Routers Basic configuration =====

CORE-R1

```
en
config
hostname CORE-R1
line console 0
password cisco
login
exit

enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
```

```
do wr
```

SSH Configuration

```
ip domain name cisco.net
username admin password cisco
crypto key generate rsa
1024
line vty 0 15
login local
transport input ssh
exit
```

```
ip ssh version 2
do wr
```

CORE-R2

```
en
config

hostname CORE-R2
line console 0
password cisco
login
exit

enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
```

```
do wr
```


----- SSH Configuration -----

```
ip domain name cisco.net
username admin password cisco
crypto key generate rsa
1024
line vty 0 15
login local
transport input ssh
exit
```

```
ip ssh version 2
```

```
do wr
```

=====

IP assignment on router interfaces (Core Layer)

=====

CORE-R1

```
int gig0/0
ip address 10.10.10.1 255.255.255.252
no shutdown
exit
do wr
```

```
int gig0/1
ip address 10.10.10.5 255.255.255.252
no shutdown
exit
do wr
```

```
int se0/2/0
ip address 103.133.254.1 255.255.255.252
no shutdown
clock rate 64000
exit
do wr
```

```
int se0/2/1
ip address 103.133.254.10 255.255.255.252
no shutdown
clock rate 64000
exit
do wr
```

CORE-R2

```
int gig0/0
 ip address 10.10.10.1 255.255.255.252
 no shut
int gig0/1
 ip address 10.10.10.5 255.255.255.252
 no shut
int se0/2/0
 ip address 103.133.254.1 255.255.255.252
 clock rate 64000
 no shut
int se0/2/1
 ip address 103.133.254.10 255.255.255.252
 clock rate 64000
 no shut
do wr
```

=====

IP assignment on router interfaces (ISP_ routers)

=====

Device: ISP-1

```
int se0/3/0
 ip address 103.133.254.1 255.255.255.252
 no shutdown
exit
do wr
```

```
int se0/3/1
 ip address 103.133.254.5 255.255.255.252
 no shutdown
exit
do wr
```

Device: ISP-2

```
int se0/3/0
 ip address 103.133.254.9 255.255.255.252
 no shutdown
exit
do wr
```

```
int se0/3/1
 ip address 103.133.254.13 255.255.255.252
 no shutdown
exit
do wr
```

- **Static & Dynamic Routing**
 - OSPF routing – Dynamic route calculation across network.
 - Default routes – Direct unknown traffic to ISP.
 - Static routing – Fixed path for specific destinations.

=====

OSPF Configurations in Multilayers

=====

Device: Mlt-SW1

```
ip routing
router ospf 10
router-id 2.2.2.2
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.50.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 10.10.10.0 0.0.0.3 area 0
network 10.10.10.8 0.0.0.3 area 0
do wr
```

Device: Mlt-SW2

```
ip routing
router ospf 10
router-id 1.1.1.1
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0
network 192.168.50.0 0.0.0.255 area 0
network 192.168.60.0 0.0.0.255 area 0
network 10.10.10.4 0.0.0.3 area 0
network 10.10.10.12 0.0.0.3 area 0
do wr
```

=====

OSPF Configurations in ISP (OSPF 10 is the process ID)

=====

Device: ISP-1

```
router ospf 10
router-id 5.5.5.5
network 103.133.254.0 0.0.0.3 area 0
network 103.133.254.4 0.0.0.3 area 0
do wr
exit
```

Device: ISP-2

```
router ospf 10
router-id 6.6.6.6
network 103.133.254.8 0.0.0.3 area 0
network 103.133.254.12 0.0.0.3 area 0
```

```
do wr
exit
```

```
=====
default routes on Routers
=====
```

```
ip route 0.0.0.0 0.0.0.0 se8/2/0
ip route 0.0.0.0 0.0.0.0 se0/2/1 70
do wr
```

```
=====
Multilayer Switches
=====
```

Device: Mlt-SW1

```
int gig1/0/1
ip address 10.10.10.2 255.255.255.252
no shutdown
exit
```

```
int gig1/0/2
ip address 10.10.10.10 255.255.255.252
no shutdown
exit
```

```
ip route 0.0.0.0 0.0.0.0 gig1/0/1
ip route 0.0.0.0 0.0.0.0 gig1/0/2 70
do wr
```

```
-----
```

Device: Mlt-SW2

```
int gig1/0/1
ip address 10.10.10.6 255.255.255.252
no shutdown
exit
```

```
int gig1/0/2
ip address 10.10.10.14 255.255.255.252
no shutdown
exit
ip route 0.0.0.0 0.0.0.0 gig1/0/1
ip route 0.0.0.0 0.0.0.0 gig1/0/2 70
do wr
```

=====

Default Static route (Core Layer)

=====

Device: CORE-R1=> (If problem with configuration static routing use: ip route 0.0.0.0 0.0.0.0 se0/2/0 70)

```
ip route 0.0.0.0 0.0.0.0 se0/2/0
ip route 0.0.0.0 0.0.0.0 se0/2/1 70
```

```
do wr
```

Device: CORE-R2=> (If problem with configuration static routing use: ip route 0.0.0.0 0.0.0.0 se0/2/0 70)

```
ip route 0.0.0.0 0.0.0.0 se0/2/0
ip route 0.0.0.0 0.0.0.0 se0/2/1 70
```

```
do wr
```

Step 2: Switching Configuration

- **Basic Switch Configuration**
 - Set hostname
 - Configure console and enable passwords
 - Disable IP domain lookup
 - Set MOTD banner
 - Enable password encryption
 - Configure SSH access

=====

Multi-layer switches

=====

Device: Mlt-SW1

```
en
config
```

```
hostname Mlt-SW1
line console 0
password cisco
login
exit
```

```
enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
```

```
ip domain name cisco.net
username admin password cisco
crypto key generate rsa
1024
line vty 0 15
```

```
login local
transport input ssh
exit
```

```
ip ssh version 2
do wr
```

Device: Mlt-SW2

```
en
config
```

```
hostname Mlt-SW2
line console 0
password cisco
login
exit
```

```
enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
```

```
ip domain name cisco.net
username admin password cisco
```

```
=====
```

Switches (in Access layer)

```
=====
```

Device: Sales-SW

```
en
config t
```

```
hostname Sales-SW
banner motd #No Unauthorised Acces!!!#
no ip domain-lookup
line console 0
passw cisco
login
exit
```

```
enable password cisco
service password-encryption
exit
wr = write memory
```

```
-----
```

Device: HR-SW

```
en
config
```

```
hostname HR-SW
line console 0
```

```
password cisco
login
exit
```

```
enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
```

```
do wr
```

Device: Finance-SW

```
en
config t
```

```
hostname Finance-SW
line console 0
  password cisco
  login
exit
```

```
enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
ip domain name cisco.net
username admin password cisco
crypto key generate rsa 1024
line vty 0 15
  login local
  transport input ssh
exit
ip ssh version 2
do wr
```

Device: Admin-SW

```
en
config
```

```
hostname Admin-SW
line console 0
  password cisco
  login
exit
```

```
enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
```

```
do wr
```

Device: ICT-SW

```
en
config
```

```
hostname ICT-SW
line console 0
password cisco
login
exit
```

```
enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
```

```
do wr
```

Device: ServerRoom-SW

```
en
config
```

```
hostname ServerRoom-SW
line console 0
password cisco
login
exit
```

```
enable password cisco
no ip domain-lookup
banner motd #No Unauthorised Acces!!!#
service password-encryption
```

```
do wr
```

- **VLANs**

- Create VLANs for each department (e.g., VLAN 10: Sales, VLAN 20: HR)
- Assign access ports to VLANs
- Configure trunk ports between switches
- Create a BlackHole VLAN for unused ports

Multi-layer switches

Device: Mlt-SW1

```
int range gig1/0/3-8
switchport mode trunk
vlan 10
name Sales
vlan 20
name HR
```



```
vlan 30
name Finance
vlan 40
name Admin
vlan 50
name ICT
vlan 60
name ServerRoom
exit
```

```
int range gig1/0/1-2
no switchport
exit
do wr
```

Device: Mlt-SW2

```
int range gig1/0/3-8
switchport mode trunk
vlan 10
name Sales
vlan 20
name HR
vlan 30
name Finance
vlan 40
name Admin
vlan 50
name ICT
vlan 60
name ServerRoom
exit
int range gig1/0/1-2
no switchport
exit
do wr
```

Device: Distribution-SW

```
int range fa0/1-2
switchport mode trunk
exit
vlan 30
name Finance
vlan 99
name BlackHole
exit
```

```
int range fa0/3-24
  switchport mode access
  switchport access vlan 30
exit
int range gig0/1-2
  switchport mode access
  switchport access vlan 99
  shutdown
exit
do wr
```

=====

Switches (Access layer)

=====

Device: L3-SW

```
int range gig1/0/3-8
  switchport mode trunk
vlan 10
  name Sales
vlan 20
  name HR
vlan 30
  name Finance
vlan 40
  name Admin
vlan 50
  name ICT
vlan 60
  name ServerRoom
exit
do wr
```

Device: Sales-SW

```
int range fa0/1-2
switchport mode trunk
exit

vlan 10
name Sales
exit

int range fa0/3-24
switchport mode access
switchport access vlan 10
exit
```

```
do wr
```

Device: HR-SW

```
int range fa0/1-2
switchport mode trunk
exit
```

```
vlan 20
name HR
vlan 99
name BlackHole
exit
```

```
int range fa0/3-24
switchport mode access
switchport access vlan 20
exit
```

```
int range gig0/1-2
switchport mode access
switchport access vlan 99
exit
```

```
do wr
```

Device: Finance-SW

```
int range fa0/1-2
switchport mode trunk
exit
```

```
vlan 10
name Sales
exit
```

```
int range fa0/3-24
switchport mode access
switchport access vlan 10
exit
```

```
do wr
```

Device: Admin-SW

```
int range fa0/1-2
switchport mode trunk
exit
```

```
vlan 40
name Admin
vlan 99
name BlackHole
exit
```

```
int range fa0/3-24
switchport mode access
switchport access vlan 40
exit
```

```
int range gig0/1-2
switchport mode access
switchport access vlan 99
shutdown
exit
```

```
do wr
```

Device: ICT-SW

```
int range fa0/1-2
switchport mode trunk
exit
```

```
vlan 50
name ICT
vlan 99
name BlackHole
exit
```

```
int range fa0/3-24
switchport mode access
switchport access vlan 50
exit
```

```
int range gig0/1-2
switchport mode access
switchport access vlan 99
shutdown
exit
```

```
do wr
```

Device: ServerRoom-SW

```
int range fa0/1-2
switchport mode trunk
exit
```

```
vlan 60
name ServerRoom
vlan 99
name BlackHole
exit
```

```
int range fa0/3-24
switchport mode access
switchport access vlan 60
exit
```

```
int range gig0/1-2
switchport mode access
switchport access vlan 99
```

```
shutdown
exit
```

```
do wr
```

Step 3: Inter-VLAN Routing

- **Layer-3 Switching with SVIs**
 - Interface VLAN creation – Enables L3 routing per VLAN.
 - IP assignment – Assigns default gateway per VLAN.
 - DHCP helper – Forwards DHCP requests to server.

Device: Mlt-SW1

```
int vlan 10
  no shutdown
  ip address 192.168.10.1 255.255.255.0
  ip helper-address 192.168.60.2
exit
int vlan 20
  no shutdown
  ip address 192.168.20.1 255.255.255.0
  ip helper-address 192.168.60.2
exit
int vlan 30
  no shutdown
  ip address 192.168.30.1 255.255.255.0
  ip helper-address 192.168.60.2
exit
int vlan 40
  no shutdown
  ip address 192.168.40.1 255.255.255.0
  ip helper-address 192.168.60.2
exit
int vlan 50
  no shutdown
  ip address 192.168.50.1 255.255.255.0
  ip helper-address 192.168.60.2
exit
int vlan 60
  no shutdown
  ip address 192.168.60.1 255.255.255.0
exit
do wr
```

Device: Mlt-SW2

```
int vlan 10
  no shutdown
  ip address 192.168.10.1 255.255.255.0
  ip helper-address 192.168.60.2
```

exit

```
int vlan 20
no shutdown
ip address 192.168.20.1 255.255.255.0
ip helper-address 192.168.60.2
exit
```

```
int vlan 30
no shutdown
ip address 192.168.30.1 255.255.255.0
ip helper-address 192.168.60.2
exit
```

```
int vlan 40
no shutdown
ip address 192.168.40.1 255.255.255.0
ip helper-address 192.168.60.2
exit
```

```
int vlan 50
no shutdown
ip address 192.168.50.1 255.255.255.0
ip helper-address 192.168.60.2
exit
```

```
int vlan 60
no shutdown
ip address 192.168.60.1 255.255.255.0
exit
```

do wr

- **Subnetting**

- Subnet VLANs – Allocates separate IP ranges for each VLAN to avoid overlap.
- Organized IP management – Facilitates growth and network efficiency.

Step 4: Security Measures

- **Access Control Lists (ACLs)**

- Deny Marketing (VLAN 10) from reaching Finance (VLAN 30) to ask money , and allow everything else.

Device: Mlt-SW1

```
ip access-list extended NO_MONEY
  deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
  permit ip any any
exit
interface vlan 10
  ip access-group NO_MONEY in
exit
do write memory
```

- **NAT & PAT**

- Translate private IPs to public IPs.
- Enable multiple devices to share one public IP.
- Support internal network connectivity to the internet.

=====

IP assignment on router interfaces (Core Layer)

=====

Device: CORE-R1

```
ip nat inside source list 1 int se0/2/0 overload
ip nat inside source list 1 int se0/2/1 overload
```

```
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.40.0 0.0.0.255
access-list 1 permit 192.168.50.0 0.0.0.255
access-list 1 permit 192.168.60.0 0.0.0.255
```

```
int range gig0/0-1
ip nat inside
exit
```

```
int se0/2/0
ip nat outside
int se0/2/1
ip nat outside
exit
```

```
do wr
```

Device: CORE-R2

```
ip nat inside source list 1 int se0/2/0 overload
ip nat inside source list 1 int se0/2/1 overload
```

```
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 1 permit 192.168.20.0 0.0.0.255
access-list 1 permit 192.168.30.0 0.0.0.255
access-list 1 permit 192.168.40.0 0.0.0.255
access-list 1 permit 192.168.50.0 0.0.0.255
access-list 1 permit 192.168.60.0 0.0.0.255
```

```
int range gig0/0-1
ip nat inside
exit
```

```
int se0/2/0
ip nat outside
int se0/2/1
ip nat outside
exit
```

```
do wr
-----
```

- **Port Security**

- Limit MAC addresses – Restrict number of devices per port.
- Sticky MAC – Auto-learns MAC addresses.
- Violation shutdown – Shuts port if security violated.

SW-1

```
en
config t
int f0/1
switchport mode access
switchport port-security
switchport port-security maximum 2
switchport port-security violation restrict
switchport port-security mac-address sticky
do wr
-----
```

Device: Finance-SW

```
int range fa0/3-24
Switchport port-security
switchport port-security maximum 1      //{ Means 1 user at a time}
switchport port-security mac-address sticky
switchport port-security violation shutdown
exit
```

```
int fa0/6
no Switchport port-security
no switchport port-security maximum 1
no switchport port-security mac-address sticky
```



```
no switchport port-security violation shutdown
exit
```

Step 5: Quality of Service (QoS)

- **QoS Configuration**
 - Prioritize voice/video traffic over normal data.
 - Ensure stable performance during congestion.
 - Reduce latency for real-time applications.

CORE-R1

```
en
config t
class-map match-any VOICE
  match ip dscp ef
policy-map QoS-POLICY
  class VOICE
    priority 1000
  class class-default
    fair-queue
int g0/0
service-policy output QoS-POLICY
do wr
```

Step 6: Monitoring and Management

- **SNMP Configuration**
 - Enable remote monitoring of devices.
 - Allow SNMP managers to poll data.
 - Support centralized network management.

CORE-R2

```
en
config t
snmp-server community public RO
snmp-server community private RW
snmp-server location DataCenter
snmp-server contact admin@net.local
do wr
```

- **Logging & Alerts**
 - Send system logs to a central server.
 - Enable timestamping of log messages.
 - Help in troubleshooting and auditing.

CORE-R1

```
en
config t
logging 192.168.20.100
logging trap warnings
```

service timestamps log datetime msec
do wr

Step 7: Testing and Validation

- **Router Verification:**

- show ip interface brief => Verify interface status and IP addresses.
- show ipv6 interface brief => Verify IPv6 interface status and addresses.
- show ip route => Check the IPv4 routing table.
- show ipv6 route => Check the IPv6 routing table.
- show ip ospf neighbor => Verify OSPFv2 neighbor relationships.
- show ipv6 ospf neighbor => Verify OSPFv3 neighbor relationships.
- show ppp all => ! Verify PPP link status.
- show ip nat translations => Check NAT translations.
- show access-lists => Display configured ACLs.
- ping <destination ip> => Test Layer 3 connectivity.
- traceroute <destination ip> => Trace the path to a destination

- **Switch Verification:**

- show vlan brief => Verify VLAN creation and port assignment.
- show interfaces trunk => Check trunking status.
- show etherchannel summary => Verify EtherChannel status.
- show port-security interface <interface id> => Check port security status.
- show ip interface brief => Check the management SVI status.

- **End-to-End Tests:**

- ping 8.8.8.8 source vlan 10
- traceroute 2001:4860:4860::8888

- **Troubleshooting**

- Use commands like ping, traceroute, show ip route.
 - Verify ACLs and NAT translations.
 - Check VLAN assignments and trunking.
 - ping 192.168.50.14
 - tracet 103.133.254.13
 - ipconfig /renew (from PCs)
-

Step 8: Results and Evaluation (Verifications)

- **Performance Metrics**

- Measure latency, throughput, and jitter.
- Verify packet delivery across all networks.
- Confirm QoS works for priority traffic.

- **Achievement of Objectives**

- Secure access with passwords, SSH, and ACLs.
- Enable VLANs and inter-VLAN routing.
- Provide NAT, PAT, and DHCP for dynamic IPs.
- Confirm stable monitoring with SNMP and logging.